

Sumith Chandra Satkuri

Software Engineer

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SUMMARY

Software Engineer with strong systems fundamentals and full-stack range — from low-level C (compilers, network protocols) to modern React/TypeScript and Python backends. Currently modernizing legacy enterprise applications at Insurity; previously optimized on-device systems at Samsung R&D. IIT Kharagpur CS graduate and Codeforces Candidate Master with a focus on clean, maintainable, well-architected code.

SKILLS

- **Languages:** C++, C, Python, Go, TypeScript, JavaScript, SQL
- **Backend & Systems:** FastAPI, Node.js, Express, Flask, REST APIs, POSIX/pthreads, socket programming
- **Frontend:** React, TypeScript, HTML/CSS
- **Databases & Tools:** PostgreSQL, MongoDB, Git, Linux, Docker

EXPERIENCE

Software Engineer — Insurity, Noida, India Jun 2026 – Present

- Migrating a legacy ColdFusion application to a React + TypeScript front end, improving maintainability.
- Translating legacy business logic into typed, component-based React with clean, self-documenting code.
- Delivered time-boxed migration tasks with high correctness by orchestrating Claude Code and Replit in a cross-verification workflow, reviewing both outputs to optimize speed without sacrificing accuracy.

Advanced Developer Intern — Samsung R&D Institute, Bangalore, India May 2025 – Jul 2025

- Engineered PTQ and AWQ quantization pipelines that cut model size 50% and doubled inference speed.
- Reduced GPU memory usage 60% during fine-tuning via LoRA and QLoRA parameter-efficient methods.
- Enabled 4x longer-sequence training in fixed memory using DeepSpeed ZeRO parallelism and Liger-kernel.

PROJECTS

SQL Query Parser & Optimizer — C, Flex, Bison Jan 2025 - Apr 2025

github.com/sumith2425/SQL_Parser_and_Optimizer

- Built a SQL engine in C that lexes and parses queries into relational algebra trees using Flex and Bison.
- Added pushdown and join-reordering passes with cost-based selection over 25 candidate execution plans.

PaperMind — Full-Stack AI Assistant Apr 2026 - May 2026

github.com/sumith2425/PaperMind

- Built a full-stack app with a React 19 + TypeScript frontend and a FastAPI backend streaming responses via SSE.
- Designed an agentic LangGraph router across 3 routes (document RAG, web search, direct answer) with ChromaDB.

Reliable Transport Protocol over UDP — C, POSIX, pthreads Jan 2025 - Apr 2025

github.com/sumith2425/KGP-Transfer-Protocol

- Engineered a TCP-like reliable protocol over raw UDP with a custom ksocket API and sliding-window flow control.
- Implemented sequence numbers, cumulative ACKs, and retransmission; validated under 5-50% packet loss.

Road Repair & Tracking System — Full-Stack Web App (Team of 3) Jan 2024 - Apr 2024

github.com/sumith2425/Road_Repair_and_Tracking_Software

- Built a full-stack web application with a Node.js/Express backend to report and track road-repair requests.
- Implemented role-based access control for citizens, workers, and administrators.

EDUCATION

Indian Institute of Technology (IIT) Kharagpur Nov 2022 – May 2026

B.Tech (Hons.), Computer Science and Engineering • **CGPA: 8.74 / 10** • Kharagpur, India

- Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Compilers, High Performance Parallel Programming, Machine Learning, Deep learning.

ACHIEVEMENTS

- Codeforces Candidate Master — rating 1923 (handle: Sumith_chandra).
- JEE Advanced 2022 — All India Rank 883; JEE Main 2022 — All India Rank 1329.
- National Talent Search Examination (NTSE) Scholar, NCERT.